

Introduction to Computer System

Computer System:

1. Computer Architecture

- Visible to programmer
- Has impact on the logical execution
- Exp: Instruction set, number of bits, I/O mechanism, addressing technique.

2. Computer Organization

- Operational units and their interconnection.
- Exp: Control signals, interfaces between computer and peripherals, the memory technology.

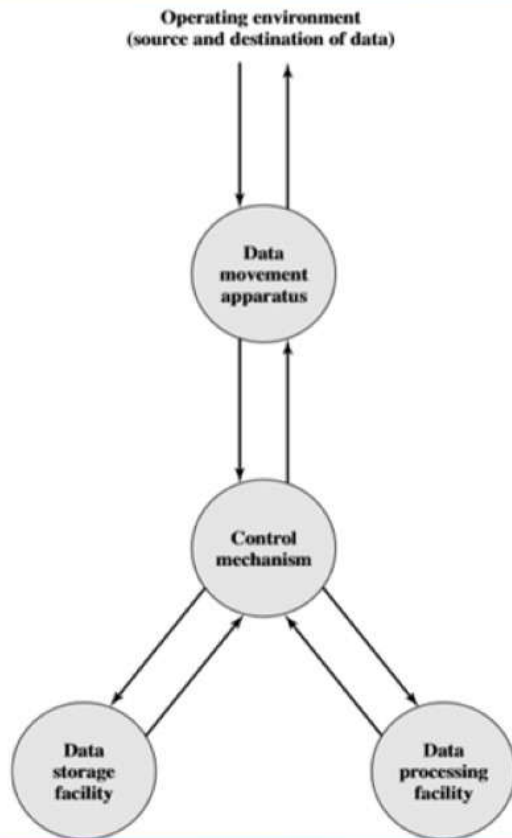
Architecture and Organization

- Same architecture but differences in organization. Thus, price and performance also different.
- Architecture may continue to be used for a long time with minor changes, but organization may change due to the technology improve.
- By changing the organization, the user can determine the system performance they required.

Structure and Function

- Computer is a complex system.
- Hierarchical nature. From the highest to lower level.
- Manage each level separately and each level will combine through interrelationships system.
- At each level:
 - **Structure** – The way the components are interrelated
 - **Function** – The operation of each individual component as part of the structure.
- **Top down** or Bottom up most effective ???

Function



i. Data Processing

- Computer used to process data.

ii. Data Storage

- Data storage either short term or long term.

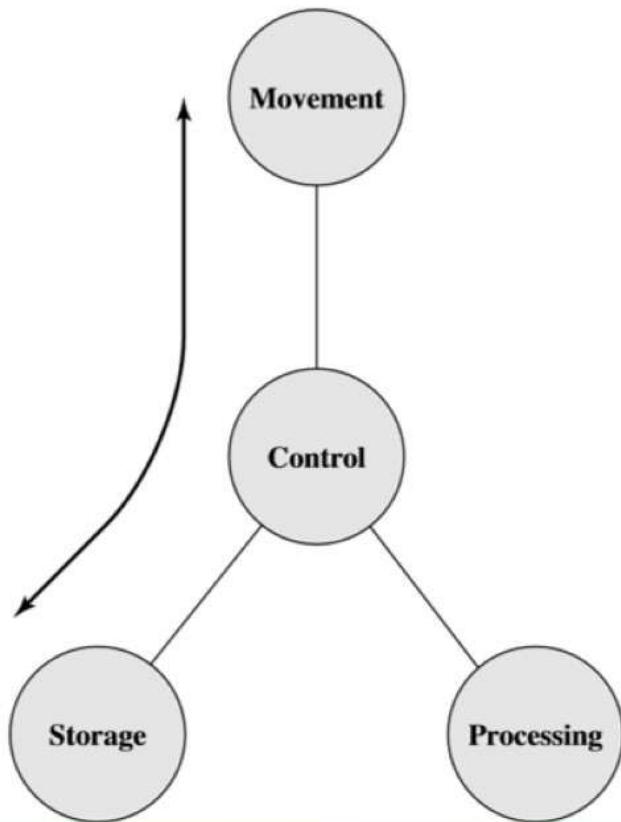
iii. Data movement

- Data communications (long distance or from a remote device)
- Input-Output (I/O), device (peripheral) directly connected to the computer.

iv. Control

- Control algorithm for the three functions above.
- User define.

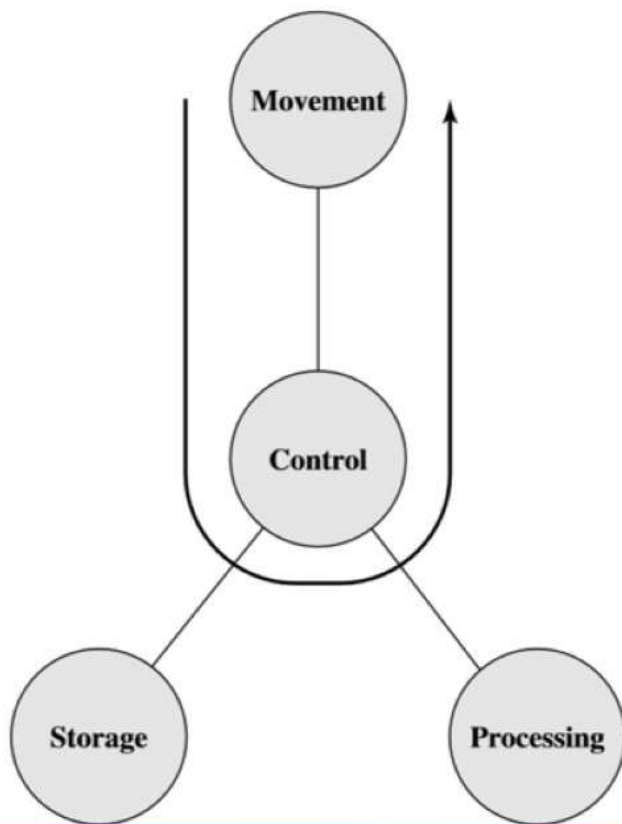
Function



Function as a **data storage device**.

- Data transferred from the external environment to computer storage (read) and vice versa (write).

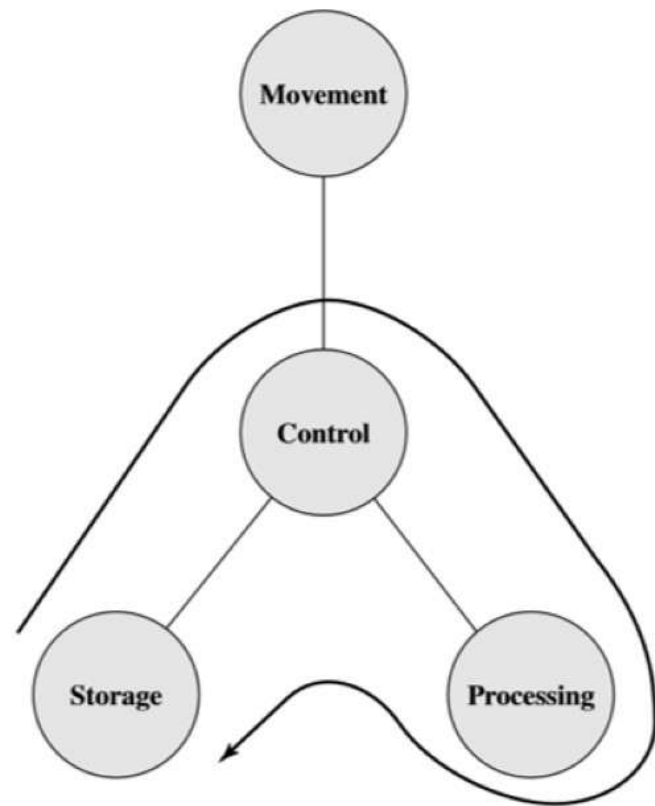
Function



Function as a **data movement device**.

- Transferring data from one peripheral or communications line to another.

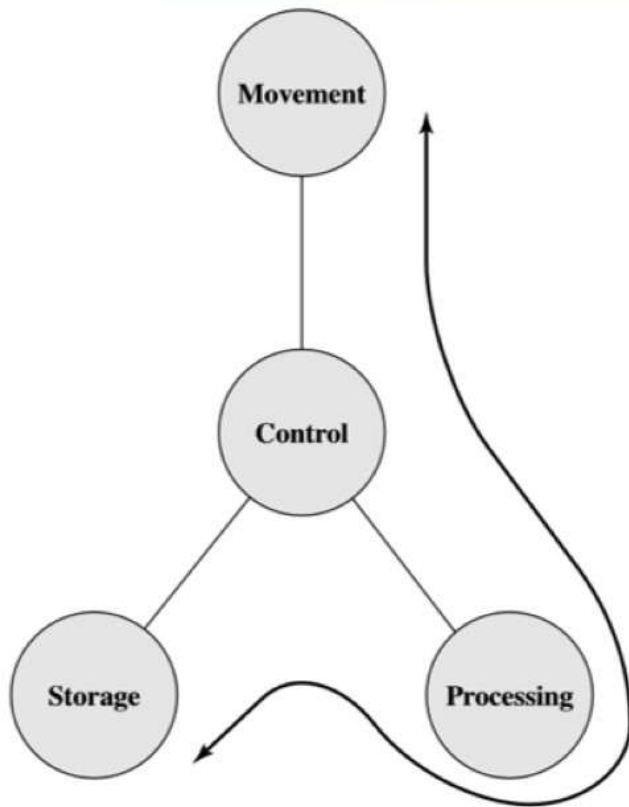
Function



Function as a **data processing**.

- Processing data in storage only.

Function



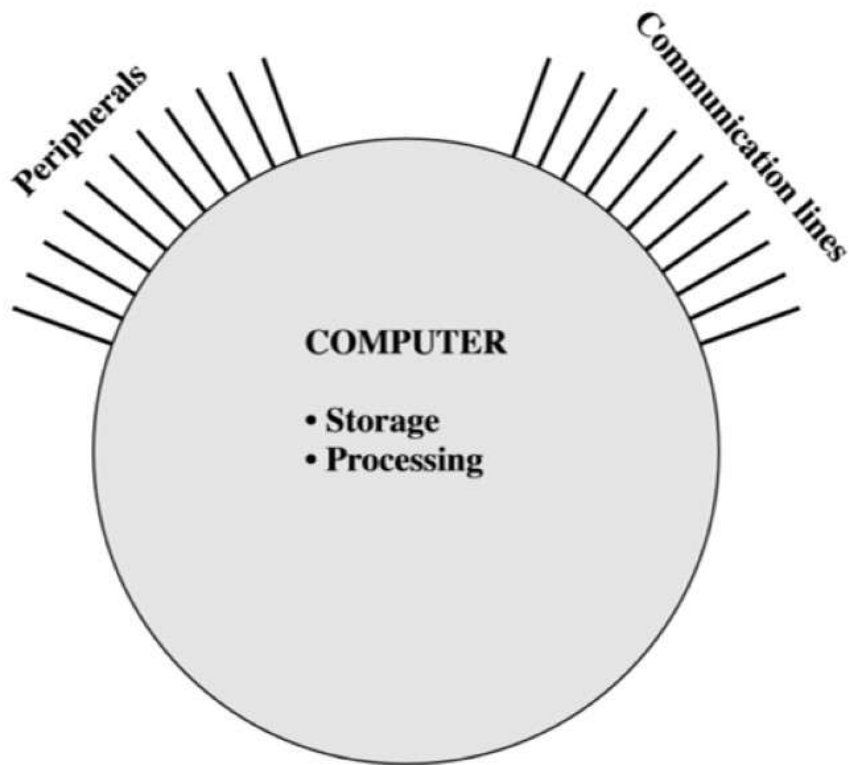
Function as a **data processing**.

- Processing data involve the storage and the external environment.

Notes:

1. All four operations involved the control function.
2. The functions can be run simultaneously with multi core processor.

Structure



- i. Computer interact with the external environment.
- ii. Peripheral devices or communication lines.

Structure (Internal Structure)

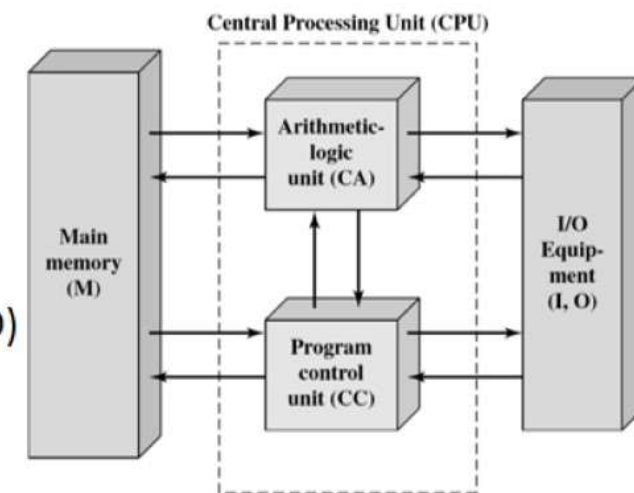
- i. **Central Processing Unit (CPU):** Controls the operation of the computer and performs its data processing functions
- ii. **Main memory:** Stores data.
- iii. **I/O:** Moves data between the computer and its external environment.
- iv. **System Interconnection:** Mechanism that provides for communication among CPU, main memory and I/O (System Bus).

Structure (CPU)

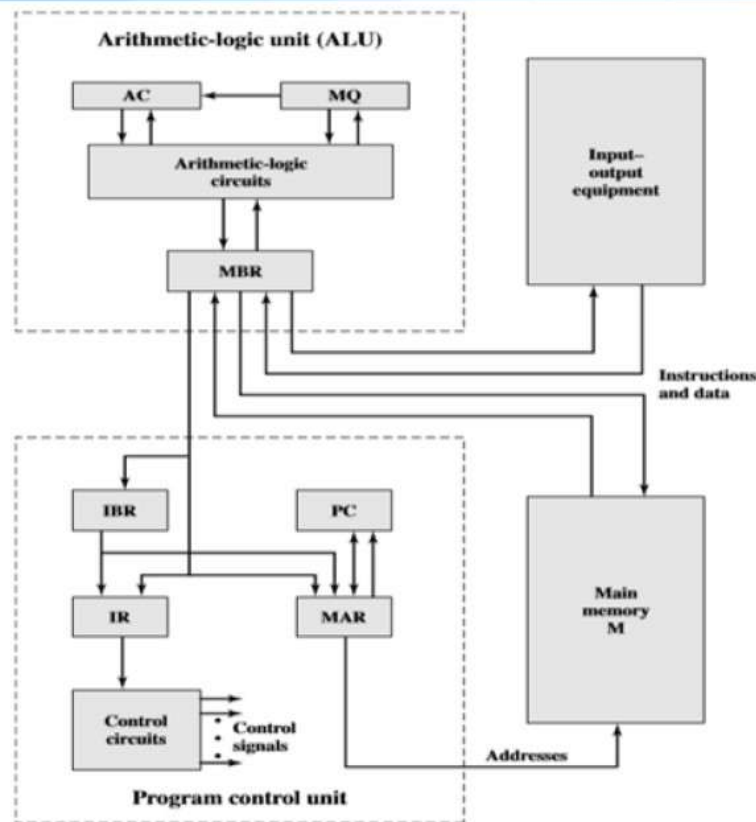
- i. **Control Unit:** Controls the operation of the CPU and hence the computer.
- ii. **Arithmetic and logic unit (ALU):** Performs the computer's data processing functions.
- iii. **Registers:** Provides storage to the CPU internally.
- iv. **CPU interconnection:** provides communication between control unit, ALU and registers.

Brief History of Computers (Vacuum tubes)

- i. ENIAC (Electronic Numerical Integrator and Computer) First general purpose computer using vacuum tubes.
 - 30 tons
 - 1500 square feet of floor space
 - 18,000 vacuum tubes.
 - 140 kW of power.
 - Decimal machine. 10 vacuum tubes represent 10 decimal digit.
 - Manually on off the switches.
- ii. EDVAC (Electronic Discrete Variable Computer)
 - Proposed by Von Neumann. (Stored-program concept)
 - Has basic internal structure of a CPU (Control unit, memory, ALU, I/O)
 - Binary system.
 - 1000 storage locations (40 bits each) for data and instructions.



Brief History of Computers (Vacuum tubes)

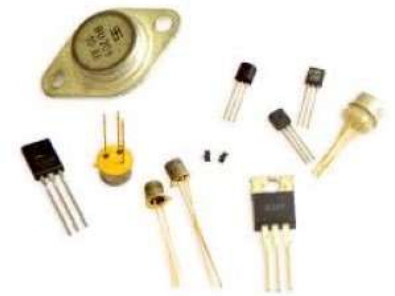


- **Memory buffer register (MBR):** Contains word to be stored in memory or sent to the I/O unit and vice versa.
- **Memory address register (MAR):** Specifies the address in memory of the word to be written from or read into the MBR
- **Instruction register (IR):** holds 8 bits opcode instruction.
- **Instruction buffer register (IBR):** hold temporarily instruction.
- **Program counter (PC):** Address of the next instruction-pair to be fetch from memory.
- **Accumulator (AC) and Multiplier quotient (MQ):** hold temporarily operands and results of ALU operations.

Brief History of Computers (Transistors)

Transistor:

- **smaller, cheaper and dissipates less heat** than a vacuum tube.
- Solid-state device, made silicon.
- IBM 7094.



Vacuum tube:

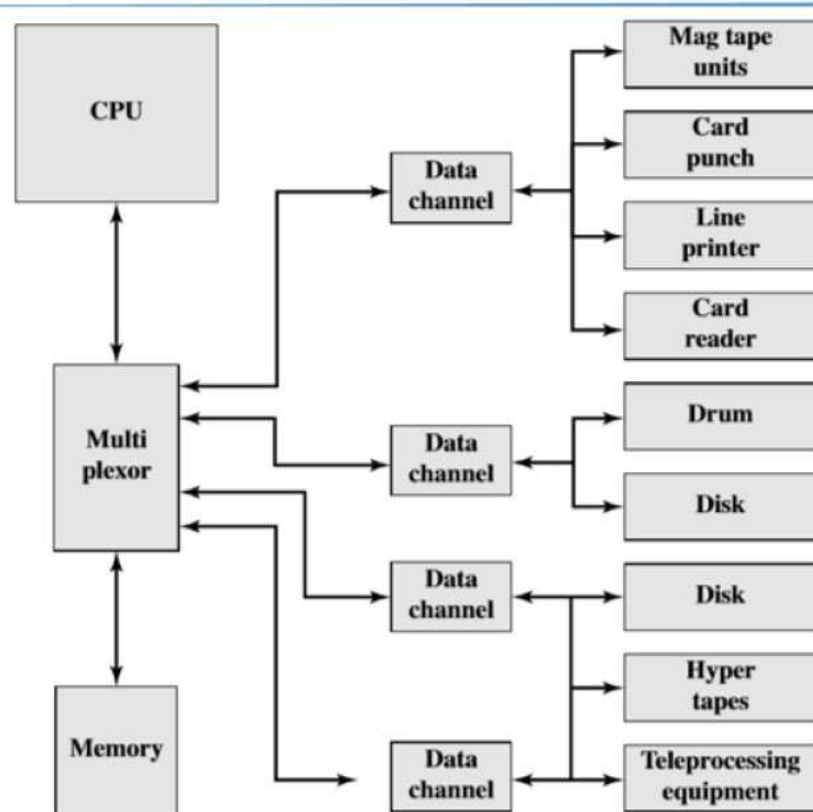
- Requires wires, metal plates, a glass capsule, and a vacuum.



Brief History of Computers (Transistors)

Model Number	First Delivery	CPU Technology	Memory Technology	Cycle Time (μ s)	Memory Size (K)	Number of Opcodes	Number of Index Registers	Hardwired Floating-Point	I/O Overlap (Channels)	Instruction Fetch Overlap	Speed (relative to 701)
701	1952	Vacuum tubes	Electrostatic tubes	30	2–4	24	0	no	no	no	1
704	1955	Vacuum tubes	Core	12	4–32	80	3	yes	no	no	2.5
709	1958	Vacuum tubes	Core	12	32	140	3	yes	yes	no	4
7090	1960	Transistor	Core	2.18	32	169	3	yes	yes	no	25
7094 I	1962	Transistor	Core	2	32	185	7	yes (double precision)	yes	yes	30
7094 II	1964	Transistor	Core	1.4	32	185	7	yes (double precision)	yes	yes	50

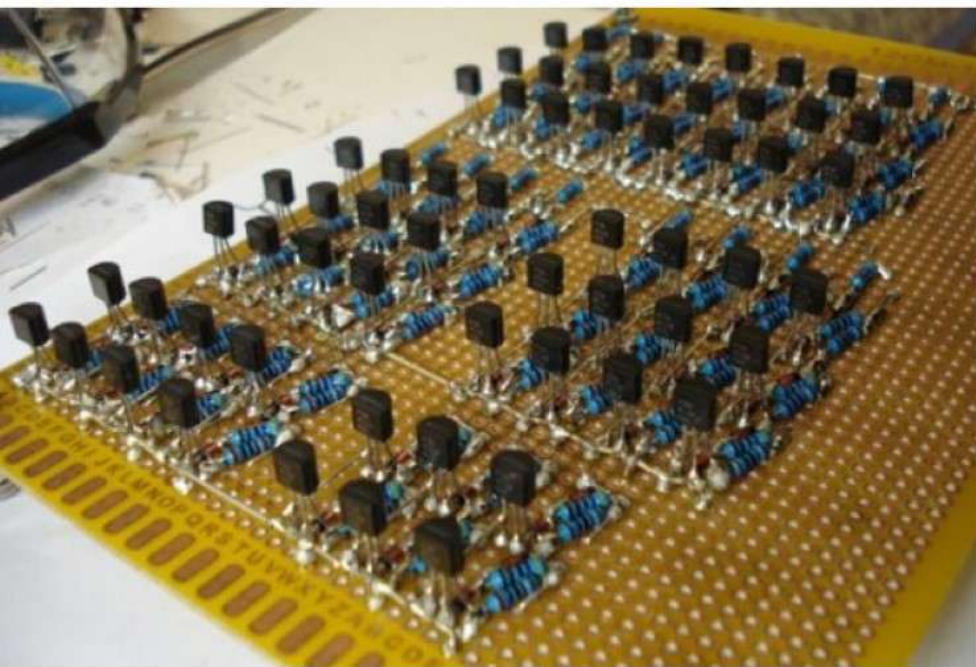
Brief History of Computers (Transistors)



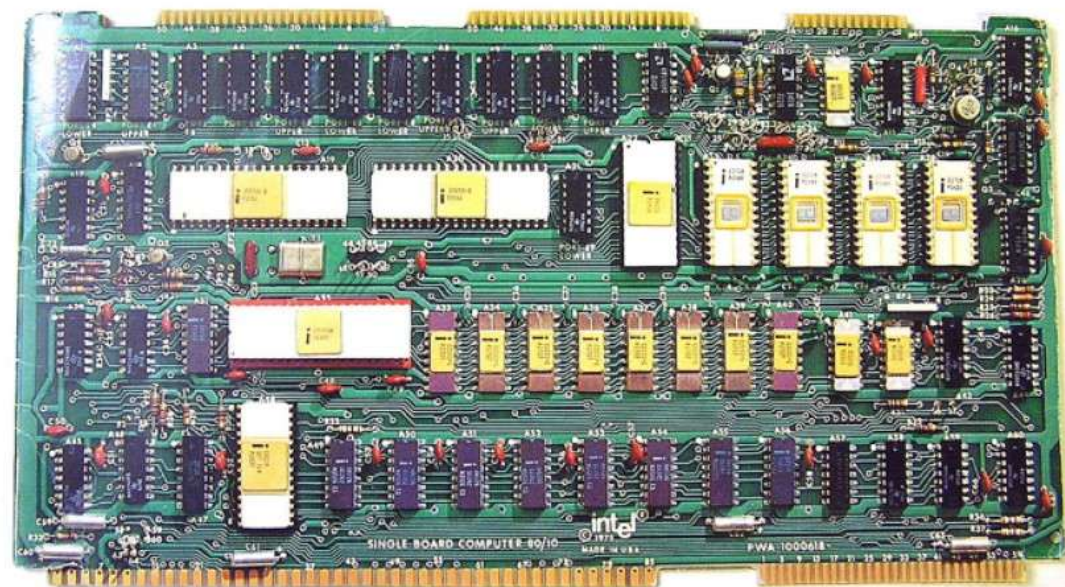
IBM 7094 Configuration:

- Main different from the IAS computer is the use of data channels.
- **Data channel** has its own processor and instruction set (I/O module).
- CPU only sent signal to the Data channel to work. Then, the system inside the data channel will complete the work and sent the data or results to CPU.
- The burden of the CPU can be reduced.
- The **Multiplexor** used to manage the data transition to CPU or memory.

Brief History of Computers (Integrated Circuits)



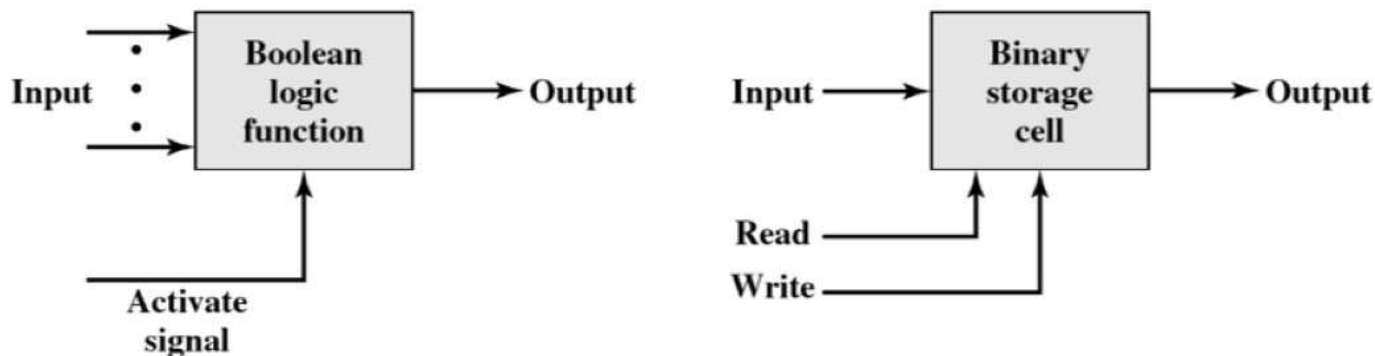
- Discrete Transistor



- Integrated Circuits

Brief History of Computers (Integrated Circuits)

- 1958, started the era of microelectronics (Integrated Circuits)
- Two elements required in the computer to perform the task which are **gates** and **memory cells**.
- Gate for **data processing** and memory cells for **data storage**.
- The paths among components used for data movement.
- Control signals used to control the gate or memory cells



Brief History of Computers (Integrated Circuits)

- The integrated circuit (transistor resistor or capacitor) can be fabricated by using the silicon wafer.
- Numbers of chip with the same configuration of **gate, memory cells and I/O connection** are fabricated in a silicon wafer.
- IBM System/360 using Integrated circuit technology, but it is not compatible with the previous version.
- However IBM System/360 is the industry's first planned family of computers.

Brief History of Computers (Integrated Circuits)

- i. Similar or identical instruction set.
- ii. Similar or identical operating system.
- iii. Increasing speed.
- iv. Increasing number of I/O ports.
- v. Increasing memory size.
- vi. Increasing cost.

Brief History of Computers (LSI)

- LSI, VLSI, ULSI (Large Scale Integration, Very-Large and Ultra-Large scale integration)
- Number of components per chip increase yearly.
- Memory:
 - In 1950 or 1960s, the memory constructed from rings of ferromagnetic material (core). Fast but bulky, expensive, and **destructive readout**.
 - 1970, Fairchild introduced **semiconductor memory**. Non-destructive and faster than core.
- Microprocessor:
 - 1971, Intel 4004: first chip to contain all of the components of a CPU on a single chip.
 - The improvement in the microprocessor follows the increase in the number of bits in the register, clock switching time and other hardware improvement.

Brief History of Computers (LSI)

- 8 Bytes vs 8 GBytes

